# 7.3 Practice: fractions (*breuken*)

← Unit 7.3 · Answers

Keep your fraction strips or some squared paper nearby. Drawing a picture is always allowed, and it's often the quickest way. Write mixed numbers like this: 2 1/2 (two and a half).

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## Set 1 · Warm-up: equal fractions and "fraction of"

1. Which is more: 1/2 or 1/4?
2. 1/2 = …/4
3. 1/2 = …/10
4. 3/4 = …/8
5. 2/3 = …/9
6. How many quarters make 1 whole?
7. 1/2 of 20 =
8. 3/4 of 20 =
9. 2/5 of 30 =
10. 1/3 of 9 =

**Talk about it:** For question 8, did you use question 7 or something else to help?

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## Set 2 · Building: simplifying (*vereenvoudigen*) and mixed numbers (*gemengde getallen*)

Make these as simple as possible:

1. 2/4
2. 5/10
3. 6/8
4. 4/12
5. 10/15
6. 12/16

Write as a mixed number:

7. 7/4
8. 11/3

Write as a fraction (no wholes):

9. 2 1/2 = …/2
10. 3 1/4 = …/4

And finally:

11. 4/4 = (as a whole number)

**Talk about it:** How do you know when a fraction is "as simple as possible"?

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## Set 3 · Building: adding and subtracting

Same denominator (*gelijknamig*) first:

1. 1/5 + 2/5 =
2. 3/8 + 4/8 =

3. 5/6 − 1/6 = (make your answer as simple as possible)
4. 3/4 + 3/4 = (write it as a mixed number)

Now different denominators (*ongelijknamig*). Make them gelijknamig first.

5. 1/2 + 1/4 =
6. 1/3 + 1/6 =
7. 3/4 − 1/2 =
8. 2/3 + 1/4 =
9. 1/2 + 1/3 =
10. 1 − 3/8 =
11. 1/2 + 1/2 =

**Talk about it:** Someone says 1/2 + 1/3 = 2/5. What would you say to them? Can you show it with a drawing?

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## Set 4 · Building: fraction × whole number, and mixed numbers

1. 2 × 1/4 =
2. 3 × 1/5 =
3. 4 × 2/3 = (as a mixed number)
4. 5 × 3/4 = (as a mixed number)
5. 1 1/2 + 2 1/2 =
6. 2 3/4 + 1 1/2 =
7. 3 1/2 − 1 1/4 =
8. 4 − 1 1/3 =
9. A cake needs 3/4 litre of milk. You bake 3 cakes. How much milk do you need?
10. Sem eats 3/8 of a pizza and Lotte eats 1/4 of it.
    a. How much of the pizza did they eat together?
    b. How much is left?
11. 2 × 1/2 =

**Talk about it:** In question 6, did you add the wholes first or the fractions first? Does it matter?

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## Set 5 · Stretch: fraction × fraction and word problems

Draw a rectangle for the first few if it helps.

1. 1/2 × 1/2 =
2. 1/2 × 3/4 =
3. 2/3 × 3/4 = (simplify)
4. 3/5 × 2/3 = (simplify)
5. 1/2 × 2 1/2 =
6. In a class of 28 children, 3/4 cycle to school. How many children is that?
7. A bottle holds 1 1/2 litres of water. Gijs drinks 1/3 of it. How much does he drink?
8. Half of the garden is a vegetable patch. A third of the vegetable patch is carrots. What fraction of the whole garden is carrots?
9. A train journey to Maastricht takes 2 1/4 hours. You've been on the train for 1 1/2 hours. How long is left? (Give your answer in hours and in minutes.)
10. 1/4 of 100 =

**Talk about it:** In question 1, you multiplied and the answer got smaller. How can that be?

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## Puzzle corner (optional)

**Ancient Egyptian fractions.** The ancient Egyptians liked to write fractions as sums of *different* unit fractions (*stambreuken*): fractions with 1 on top.

- Write 3/4 as the sum of two different unit fractions.
- Write 5/6 as the sum of two different unit fractions.
- Can you write 1 as the sum of three different unit fractions? Is there more than one way?
- Make up your own and test it on someone at home.